CENTER FOR SKILL AND ENTREPRENEURSHIP DEVELOPMENT (CSED)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.Tech Computer Science & Engineering — 2nd Year | Academic Year: 2026–2027

JOB MARKET SKILLS AND SALARY ANALYSIS

Career & Labour Market Analytics (Program Code: DA-021)

Faculty Mentor: Mr. Unwan Sir | **Team Leaders:** Harshita Upadhyay (BCS2025411) & Zainul Ali (BCS2025413)

Live Production Dashboard: https://job-market-skills-and-salary-analysis.onrender.com/
GitHub Source Code Repository:
https://github.com/upadhyayharshita07-cmd/Job-Market-Skills-and-Salary-Analysis

PROJECT TEAM ROSTER & GIT BRANCH MAPPINGS

S. No.	Student Name	Student ID	Git Branch	Primary Functional Contribution
1	**Harshita Upadhyay (TL)**	**BCS2025411**	**Harshita**	Team Leadership, System Architecture, Project Planning, Overall Coordination & Governance.
2	Lokesh Shukla	BCS2025401	Lokesh	Backend Core Development, Flask API Routing, Backend Config & Git Ignore Management.
3	Sudhanshu Singh	BCS2025402	Sudhanshu	USP Formulation, Project Research, MIT License Documentation & Uniqueness.
4	Rishabh Gupta	BCS2025406	Rishabh	Dataset Engineering, CSV Data Analysis, Dataset Structure & Schema Documentation.
5	Anurag Verma	BCS2025409	Anurag	Regex-Based Preprocessing, Text Cleaning, Data Preparation & Execution Guide.
6	**Zainul Ali (TL)**	**BCS2025413**	**Zainul**	Team Leadership, Frontend UI Development, CSS Styling, Grid & Flexbox Layout.
7	Ayushi	BCS2025414	Ayushi	Frontend DOM Manipulation, Data Binding, UI State Management & Tech Docs.
8	Nilansh Gupta	BCS2025416	Nilansh	Application Deployment, Render Hosting, Gunicorn Setup & Cache Management.
9	Pranay Kesarwani	BCS2025417	Pranay	Project Folder Structure, File Organization, Repository Structure & Documentation.
10	Mohammad Muneer	BCS2025418	Muneer	Experience Analytics, Average Experience Calculation & Future Enhancement Suggestions.
11	Kavyansh Bhadri	BCS2025419	Kavyansh	City Name Normalization, Location Tokenization, Dataset Limitations & Standardization.
12	Abhay	BCS2025421	Abhay	Full-Stack Integration, Search Functionality, Filter Implementation & Connectivity.
13	Sarvesh Singh	BCS2025422	Sarvesh	Quality Assurance, Testing, Technical Documentation & Basic Terminology.
14	Sachin	BCS2025424	Sachin	Salary Ethics Audit, Salary Boundary Verification, Data Accuracy & Interpretation.

1. Executive Abstract & Problem Statement

Undergraduate engineering students frequently struggle to identify which technical skills are most relevant to current industry demands. Informal guidance often reflects personal biases or outdated academic syllabi. Furthermore, online recruitment portals contain noisy abbreviations, multi-city listings, and unadvertised salary figures. The **Job Market Skills and Salary Analysis** project addresses this challenge through **Data-Driven Career Intelligence**, analyzing **12,000 real job postings** from Naukri across India to provide empirical insights into skill demand, role distributions, and location trends.

2. Key Project Objectives

• **Automated Preprocessing:** Ingest, sanitize, and deduplicate 12,000 job advertisements.
• **Skill Standardization:** Consolidate variants (Python, SQL, AWS, Machine Learning) using deterministic alias dictionaries.
• **Entity & City Normalization:** Resolve city aliases (e.g. Bangalore/Bengaluru -> Bengaluru) and create regional clusters (Delhi NCR).
• **Role Taxonomy:** Classify diverse job titles into 18 established industry families.
• **Experience Modeling:** Extract minimum and maximum experience thresholds using regular expressions.
• **Salary Transparency:** Explicitly declare missing salary data rather than synthesizing misleading predictions.
• **Cloud Deployment:** Deploy a high-performance Flask + Gunicorn web service on Render with sub-15ms response latency.

3. System Architecture & Methodology

The application employs a decoupled client-server architecture. The backend is built using **Flask 3.x** and **Pandas 2.x**, served via **Gunicorn WSGI** on the **Render Cloud Platform**. Data loading is secured with an in-memory **LRU Cache** (@lru_cache) ensuring lightning-fast REST responses (<15ms). The presentation tier employs semantic HTML5, modern CSS Grid/Flexbox, and asynchronous vanilla JavaScript (ES6+), eliminating unnecessary bundling overhead.

4. Empirical Findings & Quantitative Results

Analytical Dimension	Key Metric / Finding	Career & Engineering Insight
Dataset Volume	12,000 Records (5 Dimensions)	Large-scale sample of the Indian analytics and software ecosystem.
Top In-Demand Skill	Python (>1,500 mentions)	Foundational language required across ML, Data Engineering, and Analysis.
Secondary Skills	SQL, ML, AWS, Excel, Power BI	High industry emphasis on database querying, cloud deployment, and BI.
Leading Tech Hub	Bengaluru (5,093 mentions)	Accounts for ~42% of all localized technical job listings in India.
Secondary Hubs	Pune (2,014), Hyderabad (1,940)	Strong regional hiring demand in western and southern technology corridors.
Prominent Role Family	Data Engineer (~27% of dataset)	Substantial enterprise requirement for scalable ETL and cloud pipelines.
Experience Requirement	Min: 2.8 yrs | Max: 6.4 yrs	Entry-level roles (0-2 yrs) represent ~18%, requiring strong portfolio proof.
Salary Transparency	0% Disclosed (Explicitly Disclaimed)	Upholds academic integrity: zero fabricated or hallucinated salary metrics.

5. Main Project USP: Data-Driven Career Intelligence

Our core innovation lies in bridging the gap between raw recruitment noise and student career planning. By substituting subjective guessing with empirical metrics derived from 12,000 real job ads, the platform empowers students to build a high-ROI learning roadmap. Furthermore, our anti-hallucination policy demonstrates professional data ethics by reporting missing compensation figures honestly rather than fabricating artificial numbers.

6. Future Scope & Enhancements

• **NLP Skill Extraction:** Implement Transformer-based Named Entity Recognition (NER) on full job descriptions.
• **Resume Skill-Gap Matcher:** Provide an interactive tool allowing students to upload resumes and receive actionable skill-match percentages.
• **Automated Pipeline Ingestion:** Implement scheduled cron jobs to ingest updated weekly job listings from authorized APIs.

7. Conclusion

The **Job Market Skills and Salary Analysis** project provides an end-to-end demonstration of fullstack data engineering, robust API development, and data ethics. Completed by our 14-member student team under the guidance of Mr. Unwan Sir, the deployed platform serves as an accessible, high-performance career intelligence portal for engineering students.

Live Project URL: https://job-market-skills-and-salary-analysis.onrender.com/

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